

BACHELOR'S THESIS

Exploring Hyperbolic Embeddings of Finite Metric Spaces

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Abstract

Abstract der Thesis

Zusammenfassung

Hier die Zusammenfassung

Contents

1	Introduction	6
1.1	Overview and Motivation	6
1.2	Organization	6
2	Background	7
2.1	Hyperbolic Geometry	7
2.1.1	Riemannian Geometry	7
2.1.2	Lengths and Metrics	8
2.1.3	The Exponential Map	9
2.2	Numerical Optimization on Manifolds	9
2.2.1	First Order Optimization - Basis für RSGD	10
2.2.2	Riemannian Stochastic Gradient Descent - Algorithmus für Em- bedding der Daten	10
2.2.3	Line Search - Schneller Konvergenz für RSGD	10
3	Poincaré Disk and Metric Data Embeddings	11
3.1	The Poincaré Disk Model	11
3.2	Embedding Metric Data into the Poincaré Disk	11
4	Comparing Different Models of Hyperbolic Space	12
4.1	Existing Models of Hyperbolic Space	12
4.2	Evaluating the Models	12
5	Experiments and Discussion	13
5.1	Implementation Details	13
5.2	Evaluation on Toy and Real World Data	13
5.3	Conclusion	13

1 Introduction

1.1 Overview and Motivation

This section gives an overview about prerequisites from math and computer science (complex numbers, vector spaces, gradients, data, automatic differentiation, representation learning), followed by a motivation about embeddings, why they are useful in practice. WENIG FORMELN, FOKUS AUF MOTIVATION UND INTUITIVES VERSTÄNDNIS.

1.2 Organization

This section describes how the following thesis is organized and which problems are tackled in which chapter.

2 Background

2.1 Hyperbolic Geometry

In order to gain an understanding of the embedding space, the basics of Riemannian geometry are now explained with a focus on the concept of hyperbolic space.

2.1.1 Riemannian Geometry

Curvature

A fundamental concept used in Riemannian Geometry is curvature. Formally, the curvature of a plane curve γ with velocity $\mathbf{v} = \dot{\gamma}$ and the unit tangent $\mathbf{T}$ is defined as the derivation of the tangent with respect to the arc length

$$\kappa = \frac{d\mathbf{T}}{ds} = \frac{d\mathbf{T}/dt}{ds/dt} = \frac{1}{|\mathbf{v}|} \dot{\mathbf{T}} \quad (2.1)$$

[Mor93]. Geometrically interpreted, the curvature at a point p stands orthogonal to the tangent $\mathbf{T}$ and yields to the **oscillating circle** with radius $\frac{1}{\kappa}$.

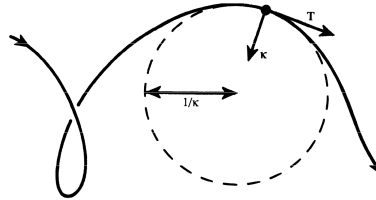
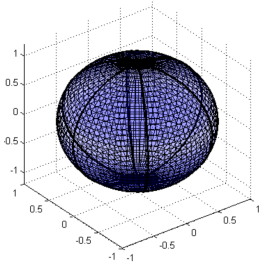
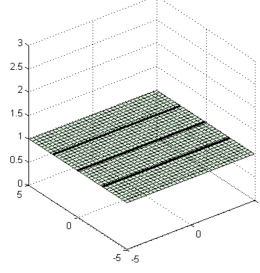
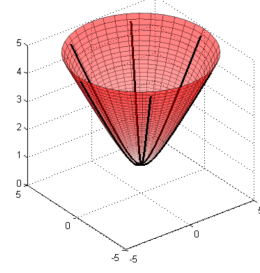


Figure 2.1: The oscillating circle of a curve [Mor93]

Surfaces in $\mathbb{R}^3$

The basis for the embedding space is a curved surface $\mathbf{S}$, which can be viewed in $\mathbb{R}^3$ (a 2-dimensional embedded submanifold $\mathbf{M}$ to be precise [Lee18]). A possibility to apply the concept of curvature to two-dimensional surfaces is the Gaussian curvature $\mathbf{K} = \kappa_1 \kappa_2$. Here κ_1 denotes the most upward (positive) and κ_2 denotes the most downward (negative) curvature. The values describing $\mathbf{S}$ are called principal curvatures.


 Figure 2.2: Sphere with $K > 0$ [Gu+18]

 Figure 2.3: Plane with $K = 0$ [Gu+18]

 Figure 2.4: Hyperboloid with $K < 0$ [Gu+18]

The embedding space used in the Poincaré Disk model is a model of hyperbolic space with constant negative Gaussian curvature.

2.1.2 Lengths and Metrics

While optimizing the embeddings, distances are computed in both the euclidean and the hyperbolic space. In the euclidean case, this is trivial using the euclidean metric, due to the zero Gaussian curvature of the space. In the hyperbolic case on the other hand, it's necessary to use some Riemannian metric, which incorporates the curvature of the space. In the derivation, the curve γ is parameterized as $\gamma = (\mathbf{u}_1, \mathbf{u}_2)$ which maps from the $\mathbf{u}_1\mathbf{u}_2$ plane to $\mathcal{S}$. The arc length of γ can then be computed as

$$L = \int_{t_0}^{t_1} |\dot{\gamma}| dt = \int_{t_0}^{t_1} \left| \left(\frac{\partial x}{\partial u_1}, \frac{\partial y}{\partial u_1}, \frac{\partial z}{\partial u_1} \right) \dot{u}_1 + \left(\frac{\partial x}{\partial u_2}, \frac{\partial y}{\partial u_2}, \frac{\partial z}{\partial u_2} \right) \dot{u}_2 \right| dt \quad (2.2)$$

$$= \int_{t_0}^{t_1} \sqrt{(\gamma_1 \cdot \gamma_1) \dot{u}_1^2 + 2(\gamma_1 \cdot \gamma_2) \dot{u}_1 \dot{u}_2 + (\gamma_2 \cdot \gamma_2) \dot{u}_2^2} dt \quad (2.3)$$

$$= \int_{t_0}^{t_1} \sqrt{\sum g_{ij} \dot{u}_i \dot{u}_j} dt \quad (2.4)$$

with $\gamma_i = \left(\frac{\partial x}{\partial u_i}, \frac{\partial y}{\partial u_i}, \frac{\partial z}{\partial u_i} \right)$ [Mor93]. In this derived formula, the term g_{ij} denotes the **first fundamental form**, which ensures that the lengths are computed correctly according to the structure of the space. This formula can be rewritten as

$$L = \int_{t_0}^{t_1} ds dt \iff ds = \sqrt{\sum g_{ij} \dot{u}_i \dot{u}_j} dt \quad (2.5)$$

so ds stands for a Riemannian metric. Using this formula, **geodesics** can be defined. The geodesic between two points corresponds to the shortest path between the points involving the local curvature of the space.

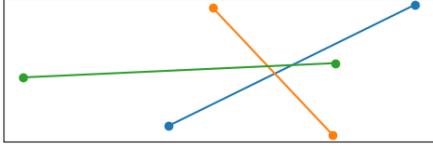


Figure 2.5: Geodesics in the euclidean plane

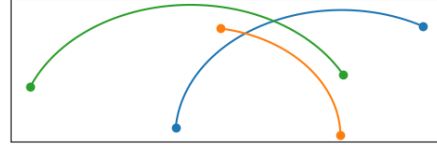


Figure 2.6: Geodesics in the hyperbolic Lobachevsky plane

2.1.3 The Exponential Map

During the optimization of the embeddings, data points in the embedding space are assigned to new points on the manifold M . The direction and length of the change depends on the Euclidean gradient ∇_E , which lies in the tangential space $T_p M$ of M at point p . This leads to a problem, as the recalculated points have to lie inside M again. The **exponential map**

$$\exp_p(v) = \gamma(|v|, p, \frac{v}{|v|}) \quad (2.6)$$

can be used to solve this [Car92]. Geometrically speaking, it assigns a point $p \in M$ together with a vector $v \in T_p M$ to a new point in M by following the geodesic γ with length $|v|$ and initial velocity $\frac{v}{|v|}$ at point p .

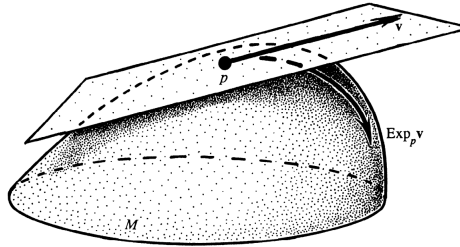


Figure 2.7: The exponential map at point p [Mor93]

2.2 Numerical Optimization on Manifolds

overall goal and procedure, Riemannian Stochastic Gradient Descent, line search, etc.

2.2.1 First Order Optimization - Basis für RSGD

2.2.2 Riemannian Stochastic Gradient Descent - Algorithmus für Embedding der Daten

2.2.3 Line Search - Schneller Konvergenz für RSGD

3 Poincaré Disk and Metric Data Embeddings

[NK17]

3.1 The Poincaré Disk Model

3.2 Embedding Metric Data into the Poincaré Disk

4 Comparing Different Models of Hyperbolic Space

4.1 Existing Models of Hyperbolic Space

Poincaré half plane, Poincaré disk, Hyperboloid model, etc.

4.2 Evaluating the Models

5 Experiments and Discussion

5.1 Implementation Details

Implementation in torch or jax with focus on scalability and hardware acceleration

5.2 Evaluation on Toy and Real World Data

Results on toy data and real world data, correctness, distortion, runtime efficiency, memory efficiency

5.3 Conclusion

What went good, what wrong, what have we learned.

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